CRICKETEC · EXPERIMENTS 4–7 (ANALYSIS EXPERIMENTS)

Camera-Perspective, Efficiency, Semantic Error & Statistics

These four experiments **analyse the five already-trained models** — no training. They reuse Experiment 1's per-clip predictions when available, so most run in seconds on CPU. All drop into the same `cricketec\` folder.

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0. Setup — one shared helper + predictions

Copy these into the `cricketec\` folder (next to `evaluate_test.py`):

- `cec_infer.py` — **required by all four** (shared model-loading + predictions helper).
- `experiment_camera_perspective.py`, `experiment_model_efficiency.py`, `experiment_semantic_error_analysis.py`, `experiment_statistical_analysis.py`.

Run Experiment 1 first (`python evaluate_test.py --device cuda`). It writes `test_eval\test_predictions.csv`, which 4/6/7 read directly — then they need no GPU and stay consistent with Exp 1. If that file is absent, each script re-runs inference itself (needs the checkpoints + GPU).

> **Resolve the Config-A/B checkpoint issue first.** In the current Exp 1 run, models A and B score far below their validation numbers (A ≈ 1.87%). Until the correct A/B checkpoints / label order are in place, every experiment's A and B rows will be wrong — C, D, E are fine. These scripts are correct; they just report whatever the checkpoints give.

4. Camera-perspective robustness

Needs your **perspective annotations** — the merged dataset JSON, or a CSV with `file_path` + `camera_perspective` (matched to clips by file name). Clips with no perspective label are **reported as missing, never guessed**.

```
python experiment_camera_perspective.py --annotations annotations\merged.json --device cuda
# or a CSV:
python experiment_camera_perspective.py --annotations annotations\manual_annotations.csv
```

Outputs in `experiment4\`: `camera_perspective_results.csv`, `camera_perspective_error_analysis.csv`, `camera_perspective_coverage.csv` (how many clips had labels), `camera_perspective_comparison.png`.

5. Computational efficiency & complexity

Profiles each model: parameters, GFLOPs, model-only latency, throughput, peak GPU memory, and frame-extraction time, paired with accuracy. **Needs a GPU for meaningful latency/memory**, and (optional) `pip install thop` for FLOPs — without thop it still reports everything else.

```
pip install thop            # optional, for GFLOPs
python experiment_model_efficiency.py --device cuda
```

Outputs in `experiment5\`: `model_efficiency_results.csv`, `accuracy_vs_latency.png`, `accuracy_vs_model_complexity.png`.

6. Semantic-layer & misclassification analysis

Per model: accuracy + macro-F1 for the 10 batting-stroke classes vs the 5 match-outcome classes, a 15×15 confusion matrix, the most-confused class pairs, and how often errors stay within a semantic group vs cross between groups.

```
python experiment_semantic_error_analysis.py --device cuda
```

Outputs in `experiment6\`: `semantic_layer_results.csv`, `class_confusion_analysis.csv`, `misclassified_clips.csv`, `semantic_group_comparison.png`, `semantic_cm_<MODEL>.png`.

7. Statistical reliability

95% confidence intervals (paired bootstrap) for accuracy and macro-F1, McNemar's exact test of the best model vs each other, Holm correction, and difference CIs — to judge whether the accuracy gaps are statistically real.

```
python experiment_statistical_analysis.py --device cuda
python experiment_statistical_analysis.py --bootstrap 5000   # more resamples
```

Outputs in `experiment7\`: `statistical_analysis_results.csv`, `pairwise_significance_results.csv`, `confidence_interval_comparison.png`.

> These tests quantify uncertainty on the **existing test partition** — they do not measure variability across training seeds. Reporting mean ± SD across seeds would require retraining with multiple seeds.

8. If something goes wrong

Problem	Fix
`ModuleNotFoundError: cec_infer`	`cec_infer.py` isn't in the folder — copy it next to the experiment scripts.
Exp 5 GFLOPs column blank	`pip install thop` (optional); everything else still reports.
Exp 4: "No grouped results"	The annotation clip names don't match the test video names, or `camera_perspective` is empty — check the coverage CSV.
Slow / re-running inference	Run Experiment 1 first so `test_eval\test_predictions.csv` exists; 4/6/7 then reuse it with no GPU.
A & B results look wrong	Expected until the A/B checkpoint issue is resolved — see the box in section 0.